Portfolio — Design Prompt & Content Material

Areeha Zulfiqar Chaudhary · AI & Full-Stack Engineer. Part A = the design brief to paste into Figma. Part B = the finished copy to insert into the site. Anything in {{double braces}} is a placeholder to fill in.

PART A — Portfolio Design Prompt (paste into Figma / AI design tool)

Design a premium personal-portfolio website for an AI & Full-Stack Engineer. It must feel modern, elegant, premium, and clean — the kind of site that makes a recruiter instantly think "this person ships polished products." Lean on generous whitespace, refined typography, restrained colour, and subtle motion. Reference quality bar: Linear / Vercel / Stripe — crafted, not a busy template. Use the REAL content from Part B, never lorem ipsum.

Persona

- Name: Areeha Zulfiqar Chaudhary
- Role: AI & Full-Stack Engineer — LLM apps, RAG, agents
- Headline / tagline: see Part B → Hero

Art direction

- Mood: premium, calm, technical-but-human; editorial spacing and rhythm.
- Theme: clean light base (near-white #FAFAF8 background, ink #14151A text) with an optional deep-ink hero section. One accent colour {{ACCENT — suggested electric indigo #4F46E5 or emerald #10B981}}, used sparingly for links, highlights and CTAs.
- Typography: a characterful sans for headings (Sora / Geist / Clash Display) + Inter for body. Big, confident headings with tight tracking; comfortable body line-height.
- Texture: subtle only — soft shadows, 1px hairline borders, rounded-2xl cards, a faint grid or noise on the hero only. No heavy gradients.
- Motion: gentle fade / slide-up on scroll, smooth hover states, one tasteful animated accent in the hero (soft gradient blob or fine particles). Nothing gimmicky. Respect prefers-reduced-motion.

Layout — single page, sticky anchored nav

- Navbar (sticky, minimal): monogram/name left; links Work · About · Skills · Contact; a subtle "Resume" button right.
- HERO: large headline (tagline), one-line subtext, two CTAs (View Work / Get in touch), social icons (GitHub, LinkedIn, email), and a small availability pill ("Open to AI / Full-Stack roles"). Quiet visual on the right or as background.
- ABOUT: a short confident paragraph + a compact "at a glance" stat strip.
- SKILLS: grouped chips/cards — AI Engineering highlighted FIRST (accent border), then Frontend & Backend, then Data & Tools.
- FEATURED PROJECT (Roamio): full-width showcase and the visual centrepiece — title, one-liner, a horizontal architecture diagram (search → plan → re-plan loop → conditions → write), 3–4 highlight bullets, tech badges, and Live Demo + GitHub buttons. Give it the most weight on the page.
- PROJECTS GRID: cards for SmartMeet (FYP), Uraan, Volt — cover image, title, one-liner, tech badges, links. Each opens a detail view/page with: overview, problem, solution, features, stack, my role, links.
- ROLES I FIT (recruiter band): "Where I add value" — 3–4 role cards, each with a one-line why.
- CONTACT: bold closing CTA, email + socials, "Download Resume".
- Footer: minimal — name, year, quick links.

Components

- Project card, tech badge/chip, architecture diagram, stat strip, role card, primary (accent) + ghost CTA buttons, section header (small eyebrow label + big title).

Interactions

- Scroll-reveal, hover lift on cards, animated underline on links, smooth anchor scrolling, copy-email-to-clipboard, subtle parallax on the hero only.

Responsive & accessibility

- Flawless mobile: hero stacks, nav collapses to a clean menu, project grid → single column, diagrams scroll horizontally. Mobile must feel as premium as desktop.
- Strong contrast, visible focus states, semantic structure, reduced-motion support.

PART B — Portfolio Content Material (the finished copy to insert)

Hero

Name: Areeha Zulfiqar Chaudhary

Title: AI & Full-Stack Engineer

Tagline: I build AI products that are grounded, not guessed — RAG, agents, and the full-stack to ship them.

Subtext: Software engineer specialising in LLM applications — RAG, AI agents, evaluation harnesses — with production full-stack experience and four AI products built end to end.

Availability pill: Open to AI Engineer / Full-Stack roles

Primary CTAs: View Work · Get in touch

Social: GitHub · LinkedIn · Email

About

I'm Areeha — an AI & Full-Stack Engineer based in Lahore. I build AI-powered products end to end: from RAG pipelines, LLM agents, and evaluation harnesses to the Next.js / FastAPI systems that serve them. My guiding principle is simple — the LLM writes the words, deterministic code makes the decisions — so what I ship is grounded and trustworthy, not just an impressive demo. I'm completing my BS in Software Engineering at PUCIT, work part-time shipping production full-stack apps, and have built four AI projects from scratch.

At a glance: 4 AI products shipped · Full-stack (Next.js + FastAPI) · RAG & LLM agents · Computer Vision · TA for Artificial Intelligence

Skills

AI Engineering: LLM Applications (OpenAI API) · RAG · AI Agents (LangGraph) · Embeddings & Vector DBs (Chroma) · LLM-as-Judge & Evals · Structured Outputs · Prompt Engineering · LangChain · Computer Vision (MediaPipe / OpenCV)

Frontend & Backend: Next.js · React · TypeScript · Node.js · Express · FastAPI · ASP.NET Core · Tailwind CSS · WebRTC · Socket.IO · GraphQL

Data & Tools: PostgreSQL · Supabase · Prisma ORM · SQL Server · Git / GitHub · Firebase · Stripe · Cloudinary · Vercel · Neon · Render

Foundations: Full-Stack System Design · REST APIs · JWT / Auth · Agile / Scrum · CI/CD

Featured Project — Roamio · AI Trip Planner for Northern Pakistan

One-liner: A RAG + agent travel planner that's grounded, not guessed — the LLM writes words, code makes the decisions.

Tech stack: Next.js · React · Tailwind · FastAPI · LangGraph · LangChain · Chroma · OpenAI · Tavily · Supabase · Vercel · Render

Links: Live Demo {{ROAMIO_LIVE_URL}} · GitHub github.com/Areeha2004/roamio-agent

Overview

Roamio turns a few constraints (days, budget, start city, group, vibe, month) into a grounded, day-by-day itinerary — a sensible route, realistic costs, seasonal feasibility, permit notes, live road/weather conditions, and a shareable link. It's a RAG + agent system built on a curated travel corpus, not a ChatGPT wrapper.

The problem

LLM travel planners hallucinate — invented prices, impossible routes, places that don't exist. The missing ingredient is trust.

The approach

A LangGraph state machine: search → plan → (conditional re-plan loop) → live conditions → write. Every number a user would trust — which destinations, the route, the cost, the feasibility verdict — is produced by deterministic Python over the corpus; the LLM only writes prose. Core principle: the LLM writes the words, the code makes the decisions.

Key engineering

- Two RAG pipelines — semantic destination retrieval, plus a grounding-content corpus (Wikivoyage + Wikipedia + Tavily) so day notes cite real sources.
- An LLM-as-judge faithfulness guard that fact-checks each day's specific claims against its sources and softens anything unsupported.
- Natural-language trip editing — free text → typed edit operations via structured outputs, validated and applied deterministically.
- A lean evals harness — deterministic assertions on the planning math + an opt-in end-to-end pass that reads the system's own faithfulness scores.
- Live road/weather conditions via Tavily; streaming progress to the UI; save-on-share persistence with Supabase.

Project — SmartMeet · AI Meeting Analytics Platform (Final Year Project)

One-liner: An AI meeting platform that measures engagement, not just attendance — using computer vision.

Tech stack: Next.js · React · Tailwind · Node.js · Express · GraphQL · REST · PostgreSQL · Prisma · WebRTC · Socket.IO · Python · OpenCV · MediaPipe · Vercel · Neon

Links: GitHub {{SMARTMEET_REPO}}

Overview

An AI-assisted online meeting platform that improves virtual meetings by measuring participant engagement instead of relying on attendance, then auto-generates engagement reports after each meeting. Final Year Project for the BS in Software Engineering, University of the Punjab.

The problem

Attendance ≠ participation. Platforms like Zoom/Teams tell hosts who joined, not who was actually paying attention — so teachers, trainers, and hosts have no objective measure of engagement.

The solution

SmartMeet extends a real-time video platform with computer vision — face-presence detection, eye-gaze estimation, head-pose estimation, and facial-landmark analysis — combined into per-participant engagement scores and post-meeting analytics.

Key features

- Auth & roles, organisation and participant management.
- Real-time video/audio, in-meeting chat, screen sharing, and scheduling (WebRTC + Socket.IO).
- Engagement analysis: face detection, gaze estimation, head-pose tracking, automated attention scoring.
- Analytics dashboard: attendance, engagement analytics, per-participant stats, post-meeting summaries.

My contributions

- Frontend lead — built the full UI: auth pages, dashboards, the meeting interface, and responsive Tailwind components; integrated the frontend with backend APIs.
- Computer-vision contributor — researched gaze-estimation techniques, prepared/pre-processed datasets, worked with MediaPipe facial-landmark detection, and helped integrate CV outputs into the web app.

Engineering challenges

- Reliable WebRTC: signalling, peer connections, media streams, cross-device compatibility.
- CV accuracy across real-world webcams — lighting, camera quality, occlusions, face angles, latency.
- Integrating five independent systems (Next.js, Node, PostgreSQL, WebRTC, Python CV) behind clean, modular APIs.
- Performance: lightweight MediaPipe models and separating real-time comms from heavy CV processing.

Project — Uraan · AI Women-Entrepreneurship Platform

One-liner: A full-stack AI platform helping Pakistani women launch businesses with personalised, AI-driven guidance.

Tech stack: Next.js 15 · React 19 · Tailwind · Next.js API routes · PostgreSQL · Prisma · NextAuth.js · OpenAI (GPT-4o-mini) · Cloudinary · Vercel (~99% TypeScript)

Links: Live Demo uraan-ten.vercel.app · GitHub github.com/Areeha2004/Uraanv1

Overview

A full-stack AI platform that empowers Pakistani women entrepreneurs with personalised, AI-driven business guidance plus a community and marketplace to launch and grow ventures.

The problem

Women entrepreneurs in Pakistan often lack structured guidance, personalised roadmaps, and the professional networks needed to turn an idea into a viable business.

Key features

- AI-powered quiz that analyses skills and interests to recommend business ideas.
- Personalised roadmaps with timelines and resources.
- Collaboration marketplace connecting entrepreneurs with service providers.
- Community support network and profile management.

Highlights

- OpenAI GPT-4o-mini with structured prompt engineering driving the personalisation engine.
- Multi-provider auth (Credentials, Google, Facebook) via NextAuth.js.

- Type-safe data layer with Prisma over a normalised PostgreSQL schema (5+ related entities); ~99% TypeScript; live in production.

Project — Volt · Smart Energy Management App

One-liner: A full-stack app that tracks household energy per device, forecasts bills, and gives AI saving tips.

Tech stack: Next.js · TypeScript · Tailwind · Next.js API routes · Supabase · Prisma · OpenAI · Jest · Vercel (~98.5% TypeScript)

Links: Live Demo {{VOLT_LIVE_URL}} · GitHub github.com/Areeha2004/VoltKar

Overview

A full-stack smart-energy app that gives households visibility into their electricity use and actionable ways to cut bills.

Key features

- Per-device usage breakdowns and bill forecasting.
- GPT-powered saving recommendations tailored to a household's usage.
- Multi-user data isolation via row-level security.

Highlights

- TypeScript-first codebase (~98.5%) with a Jest testing setup.
- Row-level security for safe multi-tenant data; production deployment on Vercel.

Roles I'm a Strong Fit For (recruiter section)

Where I add value — pick the framing that matches the role:

- AI / LLM Application Engineer — I ship production LLM features: RAG, agents, structured outputs, evals, and faithfulness guards (see Roamio).
- Full-Stack Engineer (AI products) — I build the whole product, Next.js + FastAPI/Node, not just the model layer.
- GenAI Product Engineer — I turn raw LLM capability into trustworthy, user-facing features.
- Founding / Early-Stage Engineer — I take ambiguous problems to shipped, end-to-end products fast (four built solo or as lead).

Contact

Email: zulfiqarareeha06@gmail.com

GitHub: github.com/Areeha2004

LinkedIn: linkedin.com/in/areehazulfiqar

Location: Lahore, Pakistan

Resume: {{RESUME_PDF_LINK}}

Placeholders to fill before going live

- {{ROAMIO_LIVE_URL}} — Roamio deployed URL (or remove the Live Demo button if not deployed).
- {{SMARTMEET_REPO}} — SmartMeet GitHub link (or "private / available on request").
- {{VOLT_LIVE_URL}} — Volt deployed URL (or remove).
- {{RESUME_PDF_LINK}} — link to the hosted resume PDF.
- {{ACCENT}} — final accent colour choice (indigo #4F46E5 or emerald #10B981).
- Optional: a professional headshot, and a real screenshot/cover image per project for the cards.